I/O Table for Rockwell Allen-Bradley Projects (EAL Examination)

Table 1: Training Rig Layout - Switchboard Push Buttons and Indication lights

PB6 FAULTY COMPONENT - FAULTY COMPONENT INDICATOR (Red)	PB5 COMPONENT IN POSITION - DRILL RUNNING INDICATOR (Yellow)	PB4 MODE SELECTOR - CLAMP EXTENDED INDICATOR (Green)	PB3 STOP - CLAMP RETRACTED INDICATOR (Red)	PB2 MANUAL MODE - DRILL INDICATION (Yellow)	PB1 START - CONVEYOR RUN INDICATOR (Yellow)
note: N.C. p.b.			note: N.C. p.b.		

Table 2: I/O Table

Operator tag	**PLC I/O**	**PCB Address**	**Function / notes**
CONVEYOR RUN	Output	O:0/0	N.O. push button - right; solid light when conveyor is running
DRILLING (phase)	Output	O:0/1	N.O. push button; solid light when drill in operation
CLAMP RETRACTED	Output	O:0/2	N.C. push button; solid light to simulate the actuator
CLAMP EXTEND	Output	O:0/3	N.O. push button; solid light to simulate the actuator
DRILL RUNNING	Output	O:0/4	N.O. push button; will flashing when drill is operating
FAULTY COMPONENT	Output	O:0/5	N.C. push button - left; solid light when the component is detected as faulty
START		I:0/0	Indication light (Green) - right; to start the drilling cycle and resumes program running after being stopped
MANUAL	Input	I:0/1	Indication light (Yellow);

			cycles the manufacturing phases manually
STOP	Input	I:0/2	Indication light (Red); stops the running program
MODE (input) SELECTOR	Input	I:0/3	Indication light (Green); AUTOMATIC (default), MANUAL (next push), ENGINEERING MODE (next push), loops back to AUTOMATIC (next push);
COMPONENT IN POSITION	Input	I:0/4	Indication light (Yellow); simulates the sensor
FAULTY COMPONENT	Input	I:0/5	Indication light (Red) - left; simulates the sensor (when 4 pulses in 2 seconds are detected; 2 pulses are reserved for good parts)